

SOFTWARE REQUIREMENTS SPECIFICATION

TravelLoop — Group Trip Planning & Expense Splitting App

Version 1.0 · Updated: July 10, 2026 · References: BRD_TravelLoop v1.0, ERD (15 tables), BPMN As-Is/To-Be

1. Introduction

1.1. Purpose

This document specifies the functional and non-functional requirements of the TravelLoop system, providing the basis for design, development, and testing. It elaborates on the business requirements defined in the BRD into detailed software requirements.

1.2. Product Scope

TravelLoop is an all-in-one mobile app that lets self-organized travel groups manage an entire trip on a single platform: itinerary planning, group collaboration, expense tracking, settlement, and payment reminders.

1.3. Definitions and Terms

Term	Definition
Leader	The person who creates and coordinates the itinerary (organizer).
Member	A person invited to join a specific trip.
OCR	Optical Character Recognition — used to automatically read information from a receipt photo.
Debt Netting	A mechanism that consolidates cross-debts among multiple people into a minimal set of balances between pairs.
Trip Template	A static, public snapshot copy of a trip, used for others to browse and duplicate, fully separated from live operational data.
Event-based Gateway	A branching point in a business process that waits for one of several events to occur first (e.g., enough confirmations OR a deadline).

1.4. References

- Business Requirements Document (BRD) — TravelLoop v1.0
- BPMN As-Is and To-Be diagrams (Itinerary, Expenses)
- Entity Relationship Diagram (ERD) — 15 data tables
- UI Mockup (Figma)

2. Overall Description

2.1. Product Perspective

TravelLoop is a standalone product with no dependency on any other system. It integrates an OCR service to recognize receipts and a QR code generation service to support bank transfer payments.

2.2. Main Function Groups

- Account and role management.
- Itinerary and activity management.
- Expense management and bill splitting.
- Debt management and payments.
- Community trip templates.
- Notifications and sharing.

2.3. User Classes

User Class	Description
Leader (Organizer)	Creates the itinerary, invites members, finalizes the itinerary and expense summary; has the highest edit permissions.
Member	Views the itinerary, gives feedback, logs expenses, confirms consensus, pays off debts.

2.4. Assumptions and Constraints

- The app runs on mobile platforms (iOS/Android).
- An internet connection is required for data sync, notifications, and OCR recognition.
- The app does not process actual financial transactions — it only generates QR codes for bank transfers; payment happens outside the system.

3. Functional Requirements

3.1. Account Management

ID	Requirement Description
FR-01	The system shall allow users to sign up with email/phone and password, or log in via Google/Facebook/Apple.
FR-02	The system shall allow users to recover their password via their registered email.
FR-03	The system shall allow users to update their profile, including avatar and a fixed QR code for receiving payments.

3.2. Itinerary Management

ID	Requirement Description
FR-10	The system shall allow the Leader to create an itinerary in one of three ways: manual entry, AI-suggested itinerary, or browsing a community template.
FR-11	The system shall allow the Leader to add, edit, and delete activities within an itinerary (name, time, budget, category, location).
FR-12	The system shall allow the Leader to invite members to the itinerary; members receive the invitation and can choose to join or decline.
FR-13	The system shall allow members to comment directly on the itinerary to give feedback.
FR-14	The system shall notify the Leader when a new comment is posted, and notify members when the

	itinerary is updated.
FR-15	The system shall allow members to confirm their agreement with the itinerary.
FR-16	The system shall automatically allow the itinerary to be finalized when either condition is met first: over 50% of members confirm, OR 3 days have passed since publication.
FR-17	The system shall allow the Leader to publish an existing itinerary as a community template, as a static snapshot copy fully separated from live data (no members, expenses, or debts).
FR-18	The system shall allow other users to duplicate a published template to create their own new itinerary.

3.3. Expense Management

ID	Requirement Description
FR-20	The system shall allow users to log an expense either by manual entry or by photographing a receipt.
FR-21	When a user photographs a receipt, the system shall use OCR to automatically detect the amount and related details, and suggest a matching activity to link the expense to.
FR-22	The system shall allow users to review and edit expense details and select which members share the cost before confirming and saving.
FR-23	The system shall allow members to comment on a specific expense; the person who logged it is notified and can edit it accordingly.
FR-24	The system shall automatically update and recalculate the expense summary whenever an expense is added or changed.
FR-25	After the itinerary's end date, the system shall automatically summarize expenses and show exactly how much each member owes or is owed.
FR-26	The system shall allow members to confirm agreement with the expense summary; as with the itinerary, finalizing the summary requires either over 50% agreement OR a 3-day deadline.

3.4. Debt Management and Payments

ID	Requirement Description
FR-30	After the expense summary is finalized, the system shall automatically calculate and net cross-debts among multiple members into a minimal balance between each pair of users.
FR-31	The system shall automatically send a payment request with a bank-transfer QR code to each member who owes money.
FR-32	The system shall allow a debtor to confirm they have paid off a specific debt.
FR-33	If a debt is not confirmed as paid within 2 days, the system shall automatically send a reminder with the QR code.
FR-34	The system shall send a completion notification once all debts for a trip have been paid.

3.5. Roles and Permissions

ID	Requirement Description
FR-40	The system shall support a reusable role catalog across itineraries, each role tied to a specific set of permissions (view, edit, comment, invite members).

FR-41	The system shall assign a role to each member within the scope of a specific itinerary.
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3.6. Notifications

ID	Requirement Description
FR-50	The system shall send in-app notifications for: invitations to join, itinerary updates, new comments, when finalization conditions are met (itinerary/expense summary), and payment requests/reminders.

4. Non-Functional Requirements

Category	Requirement Description
Performance	The system shall process and display OCR recognition results within a few seconds of the user photographing a receipt.
Security	User passwords shall be stored using one-way encryption (hashing), never as plain text.
Data Privacy	Publicly shared trip templates shall not contain or allow inference of member, expense, or debt information from the original trip.
Availability	The system shall keep itinerary data synchronized in real time among all members of the same trip.
Scalability	The data architecture shall support adding new currencies, expense categories, or login methods without changing the core schema.
Usability	The interface shall support Vietnamese as the primary language, with minimal, simple input flows suited to non-technical users.
Compatibility	The app shall run reliably on both iOS and Android.

5. Data Requirements

Full details are specified in the separate ERD document. Summary of the main data structure:

Function Cluster	Main Tables
Members & Roles	ACCOUNT, ROLE, TRIP_MEMBER
Itinerary	TRIP, ACTIVITY
Expenses	EXPENSE, SPLIT_DETAIL, COMMENT
Confirmation	CONFIRMATION
Debt	DEBT, DEBT_REMINDER
System	SHARE_LINK, NOTIFICATION
Trip Templates	TRIP_TEMPLATE, ACTIVITY_TEMPLATE

Key data relationship principles: each trip has many activities and many expenses (one-to-many); one expense is split among many members via the SPLIT_DETAIL table (many-to-many); the DEBT table stores two foreign keys into ACCOUNT (the debtor and the creditor) to support debt netting; the two tables TRIP_TEMPLATE and ACTIVITY_TEMPLATE are fully separated from live operational data to protect privacy when a trip is shared publicly as a template.

6. Appendix

List of attached documents: BPMN As-Is diagram (as-is.bpmn), BPMN To-Be diagrams for Itinerary and Expenses, ERD (15 tables), UI Mockup (Figma), User Story Map (Figma).